01. BERT

Why was BERT way ahead of its time?

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Because it was a masked language model even during pre-covid days!



Source: Ramsri Goutham

CS 1671 / CS 2071
Human Language Technologies

Session 20: BERT

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October 29, 2026



School of Computing and Information

Assessments: exam

- There will be an exam in class on **Tue Nov 10**
- Covers everything in the class up to that point
- In-class, paper exam
- Closed-book except for 1 letter page of double-sided notes
- Conceptual
- No programming
- Class next Thu Nov 5 will be a review session for the exam
    - Final in-class quiz moved to Nov 5 since Nov 3 is **Election Day**

Assessments: homework

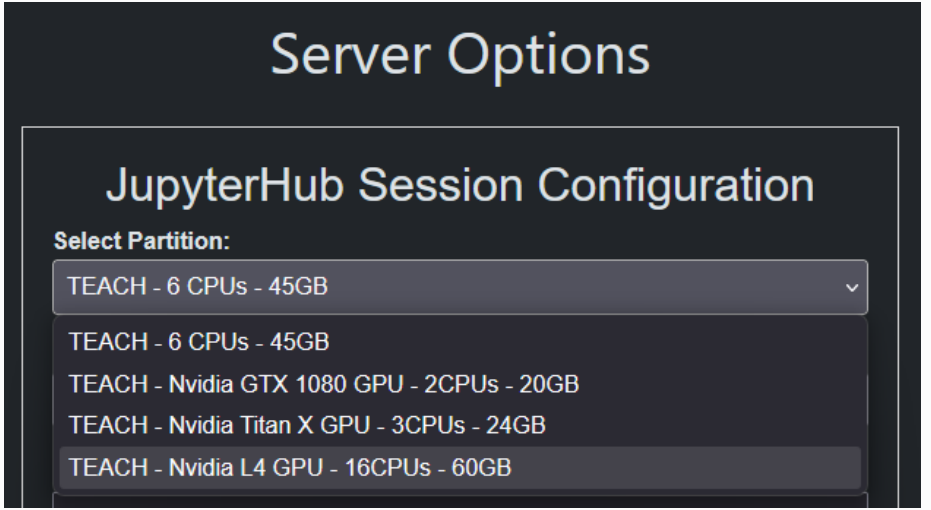
- Homework 3 has been released and is **due Nov 17**
- Run Jupyter notebooks from templates on the CRCD
- Part 1: LLM prompting
- Part 2: Instruction tuning of an LLM
  - CRCD GPUs

Overview: BERT

- Coding activity from last time: finetune GPT-2 on Shakespeare
- BERT and masked language modeling
- Encoder-decoder models: T5
- Finetuning BERT for classification
- Coding activity: finetuning BERT for text classification

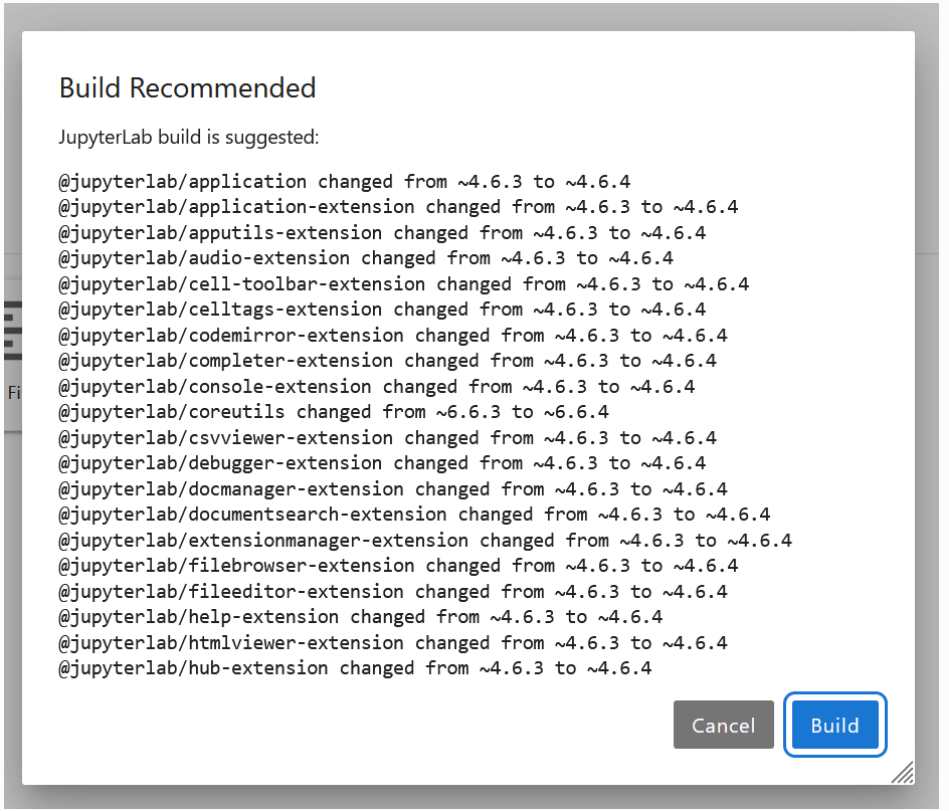
Load notebook: finetune GPT-2 on Shakespeare

1. Click this nbgitpuller link (also available on course website)
2. **Important difference from normal:** Start a server with '**TEACH – Nvidia L4 GPU – 16 CPUs – 60GB**' server
3. Load **new** custom environment at **/vast/cs1671-2026f/class-env/.pixi/envs/default**



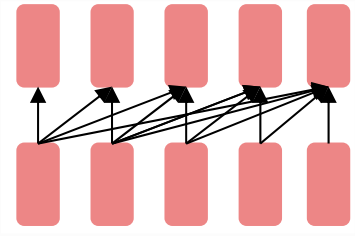
Load notebook: finetune GPT-2 on Shakespeare

4. Open **session19_gpt2_shakespeare.ipynb**
5. Push **Cancel** on Build Recommended window
6. Run the Setup cells



Review: Describe encoder, decoder, and encoder-decoder architectures

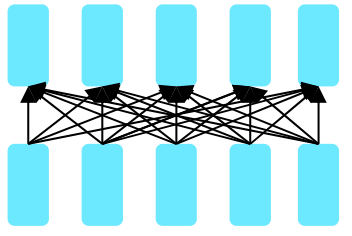
Three architectures for large language models



Decoders

GPT, Claude,

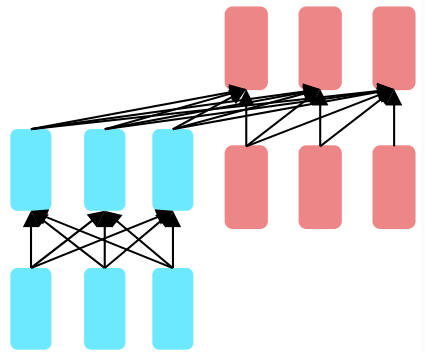
Llama, Mixtral



Encoders

BERT family,

RoBERTa



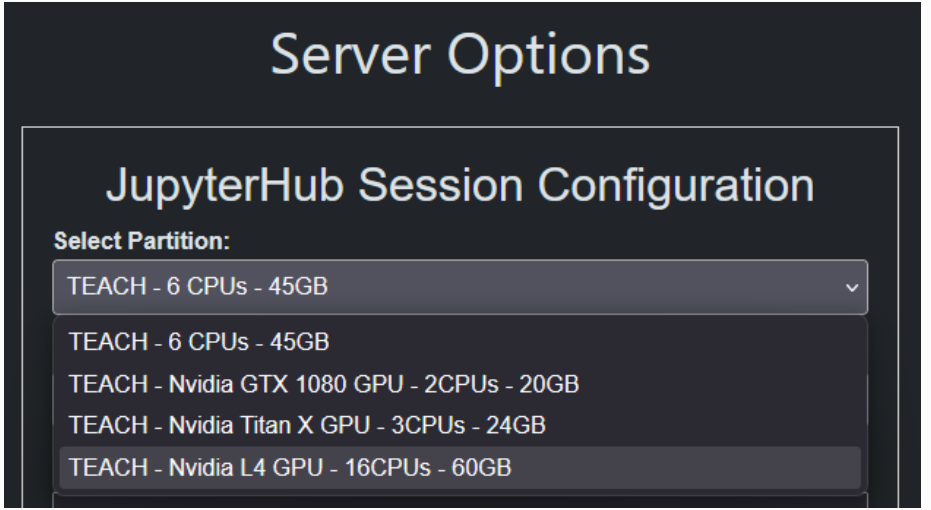
Encoder-decoders

Flan-T5, Whisper

Slide adapted from Jurafsky and Martin

Notebook: finetune GPT-2 on Shakespeare

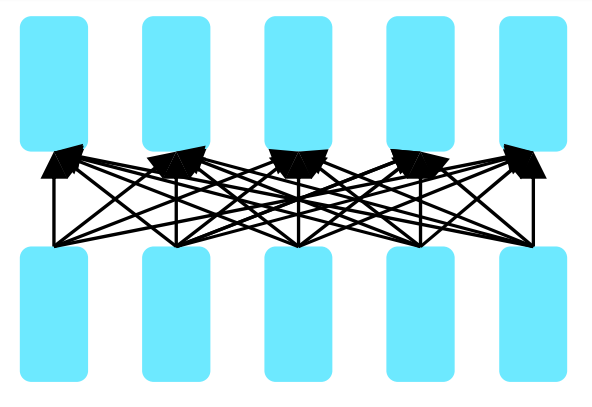
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4. Open **session19_gpt2_shakespeare.ipynb**



Transformer encoder: BERT family

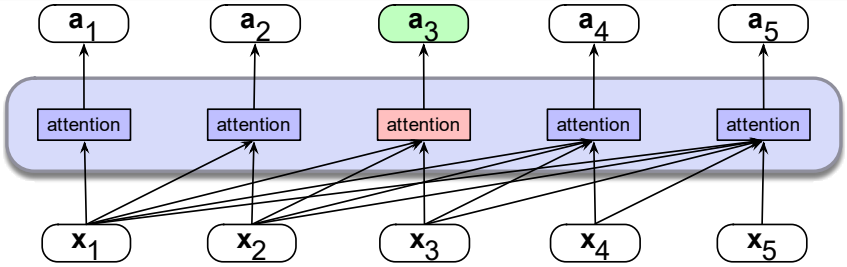
Encoders

- So far, we've looked at (causal, left-to-right) language model pretraining
- But what about tasks where we want to peek at future tokens?
- Encoders can access bidirectional context
- Map sequences of input embeddings to sequences of output embeddings that have been contextualized using information from the entire sequence
- No "masking" of future words in self-attention

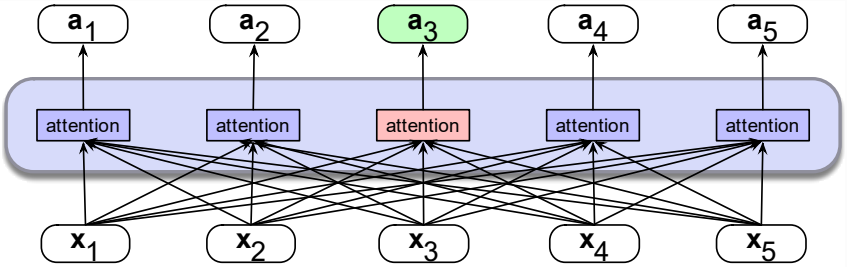


Slide adapted from John Hewitt, Jurafsky and Martin

Bidirectional Self-Attention



a) A causal self-attention layer



b) A bidirectional self-attention layer

Slide adapted from Jurafsky and Martin

Pretraining encoders: masked language modeling

- BERT (Devlin et al. 2019) is pretrained with 2 objectives
  - Masked language modeling
  - Next sentence prediction (not as important, not covered in class)

The Cloze Task

- The cloze task comes from psycholinguistics (the branch of linguistics and cognitive science that uses experimental methods to study how language works in human brains).
- It is a fill-in-the-blank task:

  He drove the yellow _____ into the front of our house.

- Subjects are presented with these frames and asked to fill in the missing words
- This allows experimenters to assess what a speaker understands about grammar, semantics, etc.
- According to the original BERT paper, this task provided the inspiration for BERT's masked language modeling (MLM) training task.
- But compare various kinds of denoising algorithms.

Slide credit: David Mortensen

MLM training in BERT

15% of the tokens are randomly chosen to be part of the masking .

Example: "Lunch was **delicious**", if delicious was randomly chosen:

Three possibilities:

1. 80%: Token is replaced with special token [MASK]

   Lunch was **delicious** -> Lunch was **[MASK]**

2. 10%: Token is replaced with a random token (sampled from unigram prob)

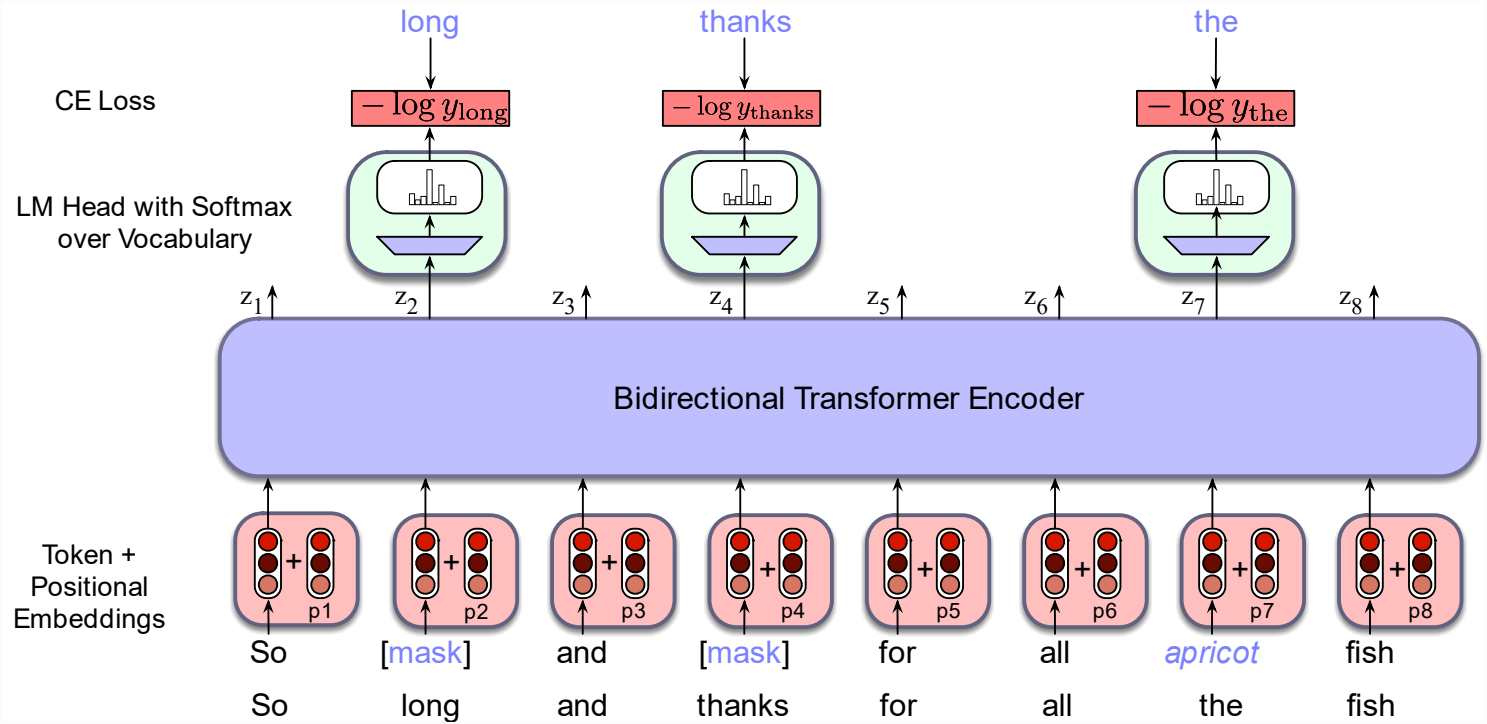
   Lunch was **delicious** -> Lunch was **gasp**

3. 10%: Token is unchanged

   Lunch was **delicious** -> Lunch was **delicious**

Slide adapted from Jurafsky and Martin

In detail



Slide adapted from Jurafsky and Martin

BERT: Bidirectional Encoder Representations from Transformers

Details about BERT

- Two models were released:
  - BERT-base: 12 layers, 768-dim hidden states, 12 attention heads, 110 million params.
  - BERT-large: 24 layers, 1024-dim hidden states, 16 attention heads, 340 million params.
- Trained on:
  - BooksCorpus (800 million words)
  - English Wikipedia (2,500 million words)
- Pretraining is expensive and impractical on a single GPU.
  - BERT was pretrained with 64 TPU chips for a total of 4 days.
    (TPUs are special tensor operation acceleration hardware)
- Finetuning is practical and common on a single GPU
  - “Pretrain once, finetune many times.”

Slide adapted from John Hewitt

Extensions of BERT

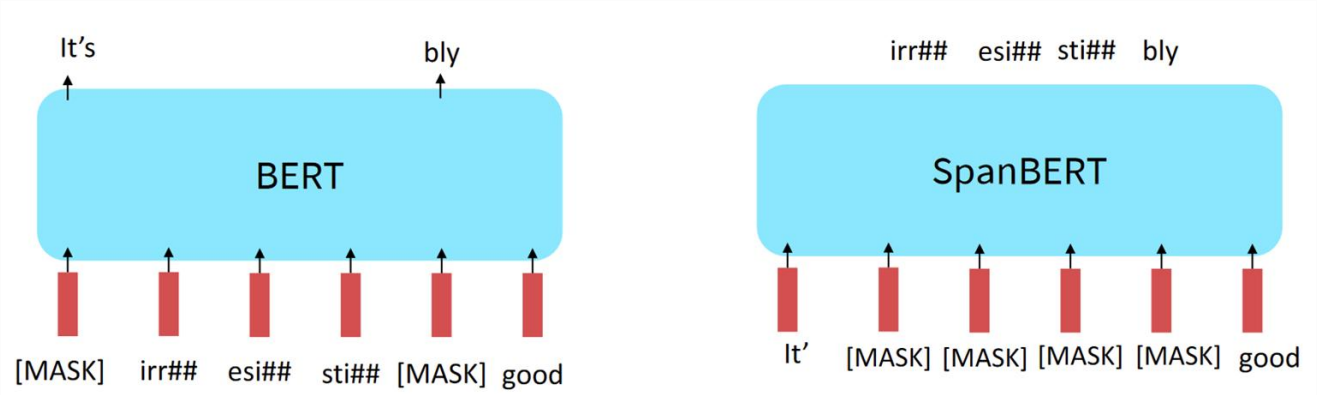
You'll see a lot of BERT variants like RoBERTa, SpanBERT, etc

Contemporary versions of BERT:

- ModernBERT
- XLM-R
- MiniLM

Some generally accepted improvements to the BERT pretraining formula:

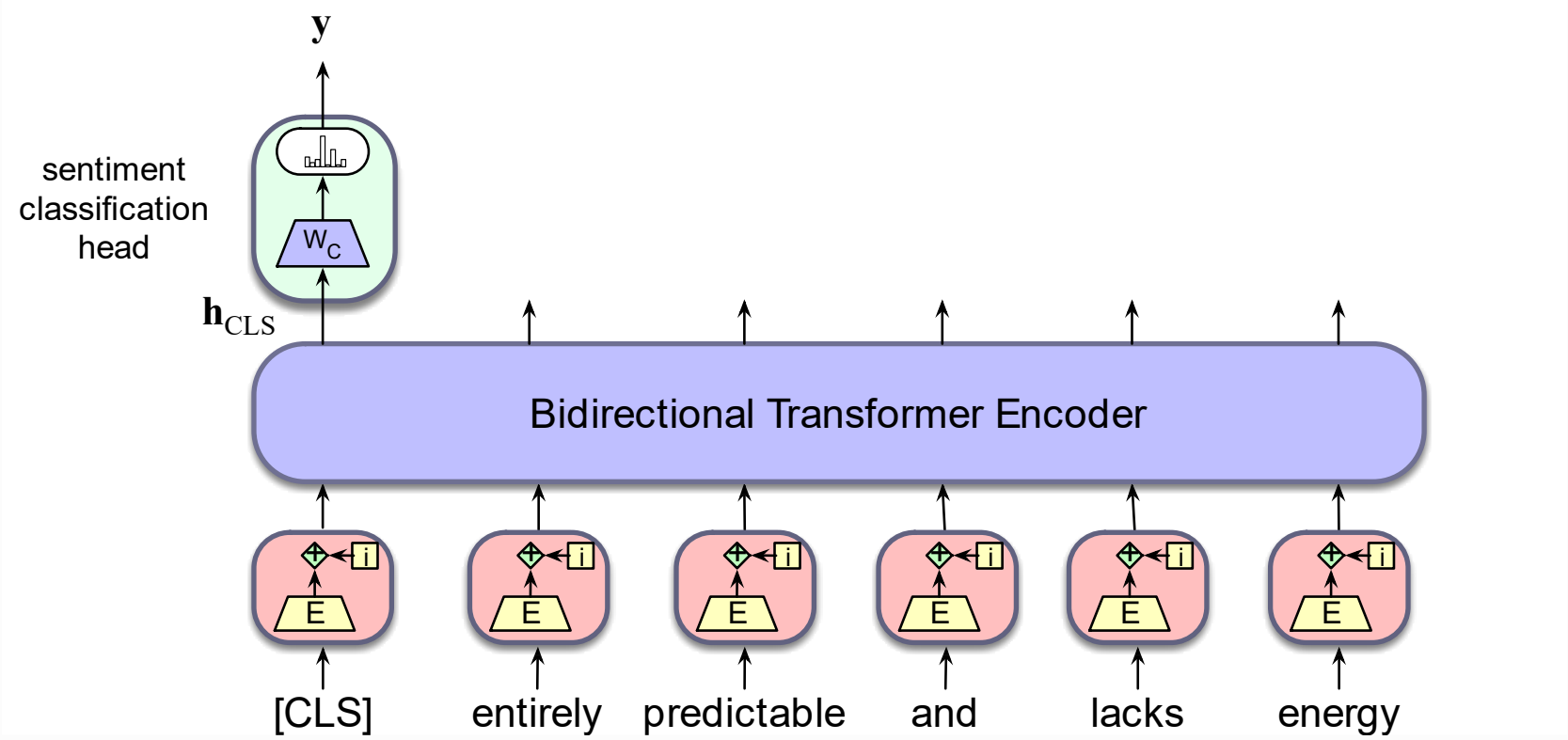
- RoBERTa [Liu et al. 2019]: mainly just train BERT for longer and remove next sentence prediction!
- SpanBERT [Joshi et al. 2020]: masking contiguous spans of words makes a harder, more useful pretraining task



Slide adapted from John Hewitt

Finetuning BERT for classification

Finetuning for classification

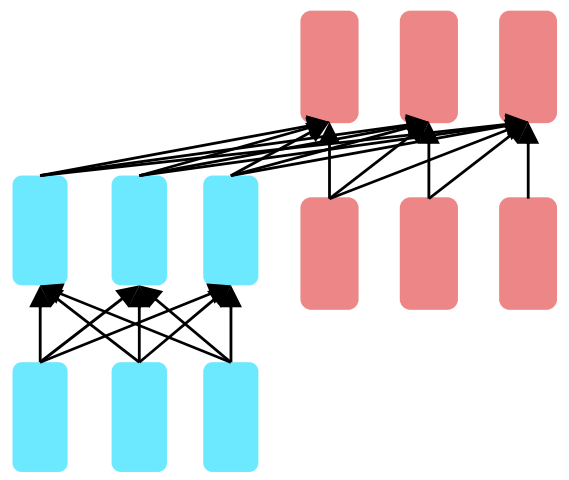


Slide adapted from Jurafsky and Martin

Encoder-decoders (T5)

Encoder-Decoders

- Trained to map from one sequence to another
- Popular for:
  - machine translation (map from one language to another)
  - speech recognition (map from acoustics to words)
- See T5 model (Raffel et al 2020)
- Decoder transformer attends to encoding



Slide adapted from Jurafsky and Martin

Conclusion

- BERT is an encoder transformer model that produces an output embedding for every input token
- BERT is pretrained on the task of masked language modeling, learning to predict masked words in the middle of sentences
- BERT is often finetuned for classification with a neural classification layer on the embedding output for the special [CLS] token (taken to represent the whole sentence)
- Encoder-decoder models like T5 map one sequence to another sequence

Notebook: finetune BERT for politeness classification

1. Click this nbgitpuller link (also available on course website)
2. **Keep using a GPU:** Start a server with '**TEACH – Nvidia L4 GPU – 16 CPUs – 60GB**' server
3. Load **new** custom environment at **/vast/cs1671-2026f/class-env/.pixi/envs/default**
4. Open **session20_bert_politeness.ipynb**

